

DEFENSE STANDARDIZATION PROGRAM

CASE STUDY

NAVSEA Test Equipment Standardization

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This case study describes how the standardization efforts of the Naval Sea Systems Command (NAVSEA) Test, Measurement and Diagnostic Equipment (TMDE) program have dramatically reduced the variety of test equipment, while driving down procurement, logistics support, maintenance, and calibration costs through modernization.

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Defense Standardization Program Case Study

NAVSEA Test Equipment Standardization

Background

The NAVSEA TMDE program buys mission-essential fleet and shore test equipment to maintain and repair more than 2,000 weapon and communication systems and equipment.

Measurement requirements were met by numerous equipment models from various commercial manufacturers. Although NAVSEA had the authority to approve and disapprove the use of models, it had no standardization criteria to prevent the proliferation of equipment with differing procurement strategies, logistics support, calibration cycles, and training.

Problem

Purchasing rules related to free and open competition resulted in many different commercially manufactured models fulfilling a single measurement requirement category. It was possible to find up to 71 different models addressing one measurement category. NAVSEA needed to eliminate the high cost of support for many different products. The constant cycle of training and retraining operators and repair technicians and the requirement to provide piece parts, technical manual, and calibration support were burdensome to the fleet users and to NAVSEA.

Moreover, many models in current inventory were obsolete. Several legacy oscilloscopes and digital multimeters had calibration intervals of fewer than 12 months. Due to their age and period of use, the equipment required more frequent calibration than originally planned, as well as more frequent corrective maintenance and repair.

Approach

In 2002, NAVSEA decided to standardize equipment with a goal of having no more than two models that satisfy each measurement requirement category. The team's strategy was to leverage commercial technology to find models meeting the measurement requirements while mitigating risk factors.

At the direction of the TMDE program manager and at the request of the representatives of fleet commanders in chief and type commanders (each of which leads a specific type of fleet element, such as the Naval Surface Force, U.S. Pacific Fleet), the TMDE team introduced an initiative that would standardize and modernize the Navy's inventory of general-purpose electrical and electronic test equipment. The basis of the plan was the phased-out replacement of obsolete, single-function, legacy test equipment with multifunction, standardized test equipment in both the operational and training environments.

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Identify Technical Requirements

With very few exceptions, all Navy electronic prime systems and equipment (PS/E) require oscilloscopes or digital multimeters for maintenance support. Therefore, these two functional groups were selected as the first candidates for the initiative.

The team identified oscilloscope measurement requirements covered by 13 categories and 71 models from a variety of manufacturers. (The oscilloscope family of equipment supports the maintenance requirements of 931 PS/E and 541 electronic work centers.) The team used a database to capture all reliability and maintenance data and calibration costs on all models for the past 5 years. The database also contains all allowance data and inventory statistics for all customers. With the database, the TMDE program can track the retirement of the legacy equipment.

The next step was to determine the engineering impact of new technology available through the commercial test equipment industry. The goal was to find a single model that could be used to cover as many requirement categories as possible with the required accuracy. The standardization effort required extensive research into PS/E measurement specifics. The team held numerous meetings and telephone conversations with in-service engineering agents and the maintenance community and conducted extensive research of available commercial test equipment capabilities.

The engineering team, headed by Naval Surface Warfare Center, Corona Division, Seal Beach Detachment, aggressively pursued market research with commercial vendors. To reduce the number of measurement requirement categories and associated models, the team looked for fewer, more capable pieces of equipment that could cover all measurement requirements needed by the fleet user. Occasionally, this approach resulted in the use of overdesigned test equipment. But by using more capable equipment across multiple measurement requirements, the program's cost savings improve dramatically and offset the slightly higher cost of the newer models.

Standardize Requirements and Purchasing

The technical team next developed purchase descriptions for items not available through the Navy stock system and reviewed the bid sample testing of proposed equipment. In an effort that spanned two fiscal years, the team reduced the number of requirement categories from 13 to 3 and the number of models from 71 to 6, covering the entire oscilloscope measurement requirement. The models selected were subjected to calibration interval analysis, resulting in a recommended 3-year calibration cycle for the new oscilloscope models.

The technical team then took the same approach to standardizing digital multimeters. The Navy reduced the number of requirement categories from 38 to 16 and the number of different models from 241 to 28.

Establish New Allowance Rules

Next, the team effort shifted to establishing new

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allowances for the more capable equipment. In general, under the old approach to establishing allowances, a piece of test equipment was allowed aboard a ship or station for shared use among 25 prime systems and equipment. The new approach uses class allowances; this approach focuses on the actual usage of the items aboard a ship or station.

NAVSEA outfits ships of the same class with a standard capability. Changes from one hull to another are normally minimal. If different ships in the same class have different prime systems and equipment, the differences are generally in the equipment manufacturer, but the measurement capability is essentially the same. This approach allows the ship to be viewed as a whole entity rather than a compilation of individual systems. The allowancing effort included numerous meetings with type commanders and fleet subject matter experts to determine the exact number of pieces of test equipment needed to support mission objectives without sacrificing capability and safety.

In the course of discussions, NAVSEA discovered that many items were required for occasional scheduled maintenance only, rather than for day-to-day ship operation. Using class allowances removed these items from the ships and placed them in a regional loan pool, dramatically reducing the amount of test equipment aboard a ship or station. Class allowancing is complete for all ship classes, as well as shore station components, for which the NAVSEA TMDE program has procurement responsibility.

Benefits

As a result of standardization, NAVSEA significantly reduced the logistics footprint of the oscilloscopes and digital multimeters. The program had been providing repair parts, repair and operator training, technical manuals, calibration procedures, and calibration standards maintenance for 312 different models. Through this effort, the number of models has been reduced to 34 across both families of equipment.

The procurement of oscilloscopes and digital multimeters was phased in over 3 years. The oscilloscope procurement started in February 2003, and the procurement of digital multimeters began in August 2003. At the time, the fleet expended $1.8 million a year for repairing and calibrating oscilloscopes and $4.3 million for maintaining and calibrating digital multimeters. The high reliability of digital phosphorus oscilloscopes resulted in the fleet saving all of the maintenance and calibration dollars in the third year of the procurement. With zero dollars expended for repair of warranty items, the fleet saved $6.1 million from reduced maintenance costs by the end of FY05. The overall savings from the reduction of the logistics tail and the decrease in calibration intervals from once a year to once every 3 years saved an estimated $45.3 million across the Future Years Defense Program. The new oscilloscopes do not need to be calibrated upon initial issue. As a result, calibration labs have been freed to concentrate on the intense, real-world deployment of ships in support of operations around the world.

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The class allowancing effort is complete through all communities and ship classes for which NAVSEA has acquisition responsibilities and is reaping real benefits in standardization and hardware footprint reduction. For the DDG-51, USS *Arleigh Burke*, this effort resulted in a reduced initial outfitting cost of more than $150,000 per ship. The effort affected 26 new construction installations for a savings of $3.9 million by FY08. Logistic support, maintenance, and calibration cost avoidance for the commissioned ships in the DDG-51 class was $2.2 million by FY05.

Class allowancing for the CV-63, USS *Kitty Hawk*, and the CVN-76, USS *Ronald Reagan*, removed more than 2,000 pieces of equipment. The equipment was used to establish regional loan pools, and the residual assets were made available to offset procurements, resulting in savings over 5 years of $5.84 million in logistic support, maintenance, and calibration cost avoidance.

Future Efforts

This initiative is being extended to address new families of test equipment, including baseband, function, and signal generators.

Lessons Learned

The following are the key lessons learned:

- *Know what you want from industry.* The key to market research is to know what you want from industry and to be steadfast in acquiring it. A request for proposals that clearly described the desired end result garnered competitive bidding and a positive, unexpected solution.
- *Prepare to change mindsets.* The class allowancing team encountered firm mindsets that were convinced that assets could not be removed from the ship for any reason. "You can't take that from me—I never know when I'll need it." When class allowancing went into effect on DDGs and administrative burdens were removed, the benefits were clear, drawing other type commanders into the near-term schedule.
- *Gain stakeholder support to ensure funding.* Funding is a challenge with any initiative. The savings realized by the TMDE program could have been significantly higher if the effort had been fully funded the first year. After the stakeholders (including 700 commands supported by the program) were briefed on initiative benefits, the program had the full support of the fleet and the funding to continue the effort.

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